Can the on-structure sensors predict inclination?

A chronological NeuralProphet experiment on same-time estimation, one-week forecasting and missing-hour reconstruction

Study report · `studies/04_neuralprophet_inclination_prediction/`

Contents

1 Introduction

Study 01 read the raw archive, decided which of the recorded values are measurements, and exported the verdict-aware archive that every later study reads. Its final inclination product — compensated for temperature, anchored once across the whole record, cleaned of impulsive noise, and carrying a flag that marks every value it replaced by interpolation — is the only inclination series used here. The intermediate versions of that channel are not used, and no value the flag marks as an interpolation is treated as a measurement. This study inherits that verdict rather than revisiting it.

The subject is one sensor at one place: the inclinometer at station 02, on the medieval wall at Porta di Sant'Ubaldo. The question is operational rather than diagnostic. Given the record as it actually stands, can the next value of the inclination be predicted — from the inclination's own history, from the other variables the instrument package on the structure measures beside it, or from external environmental proxies?

The record is not taken whole. The archive contains a 271-day outage that begins on 23 September 2022 and whose last missing day is 20 June 2023, across which nothing at all about the structure's behaviour was recorded. This study works only on what follows: the stretch running from the resumption of recording in June 2023 to the end of the archive. The choice has two reasons. That stretch is the part of the record closest to the present, and therefore the part a monitoring system deployed today would actually have to work with. It is also the only part over which the complete instrument package is available, since the wall-temperature probe and the pyranometer were installed on 21 February 2025 and record nothing before that date.

Choosing that window does not buy a continuous series. Four further whole-day outages fall inside it — 40 days from 5 May 2024, 42 days from 28 August 2024, 17 days from 26 September 2025, and 103 days from 7 March 2026 — and beneath them lies a population of shorter dropouts whose size is not yet known. Over the 27,769 hourly slots the window spans, the inclinometer returns an accepted value in 76.2% of them, and air temperature and relative humidity in 78.4%. The wall-temperature probe and the pyranometer reach only 17.7% and 18.9%: they exist for less than half the window to begin with, and they are interrupted inside the part where they do exist. What none of those numbers says is how the missing time is *distributed*: whether it arrives as isolated absent samples, as hours, or as days. Study 01 recorded the omission explicitly, noting that the anatomy of the gaps is a natural extension of it and was not attempted in it. Establishing that anatomy is the first task of this study, because the shape of the missing data decides what kind of statistical problem filling it is: recovering a scattered sample from its immediate neighbours is a different proposition from reconstructing a season.

The study therefore proceeds in four movements. It first looks at the window whole, plotting every variable the on-structure package records over it on one shared clock, with the gaps left as gaps and nothing interpolated before drawing. It then inventories those gaps, classifying the missing time by duration so that the scattered absences are separated from the outages. It then weighs the ways each class might be filled, and what it would cost to be wrong. Only then does it put the prediction question: what the next inclination value can be inferred from, using the on-structure measurements.

This updated study specifically executes this fourth movement using a refined NeuralProphet architecture. Relying on air temperature as a single predictor (after confirming a zero phase lag, $d = 0\,\mathrm{h}$, in previous analyses), the pipeline incorporates a contemporaneous nowcast alongside 12-hour lagged regressors. It maps sensitivity across horizons up to 168 hours, establishing a definitive 48-hour limit of positive skill against a seasonal naive baseline. Furthermore, it achieves rigorously calibrated uncertainty bounds using 90 % pinball quantiles and horizon-adaptive block-bootstrapping.

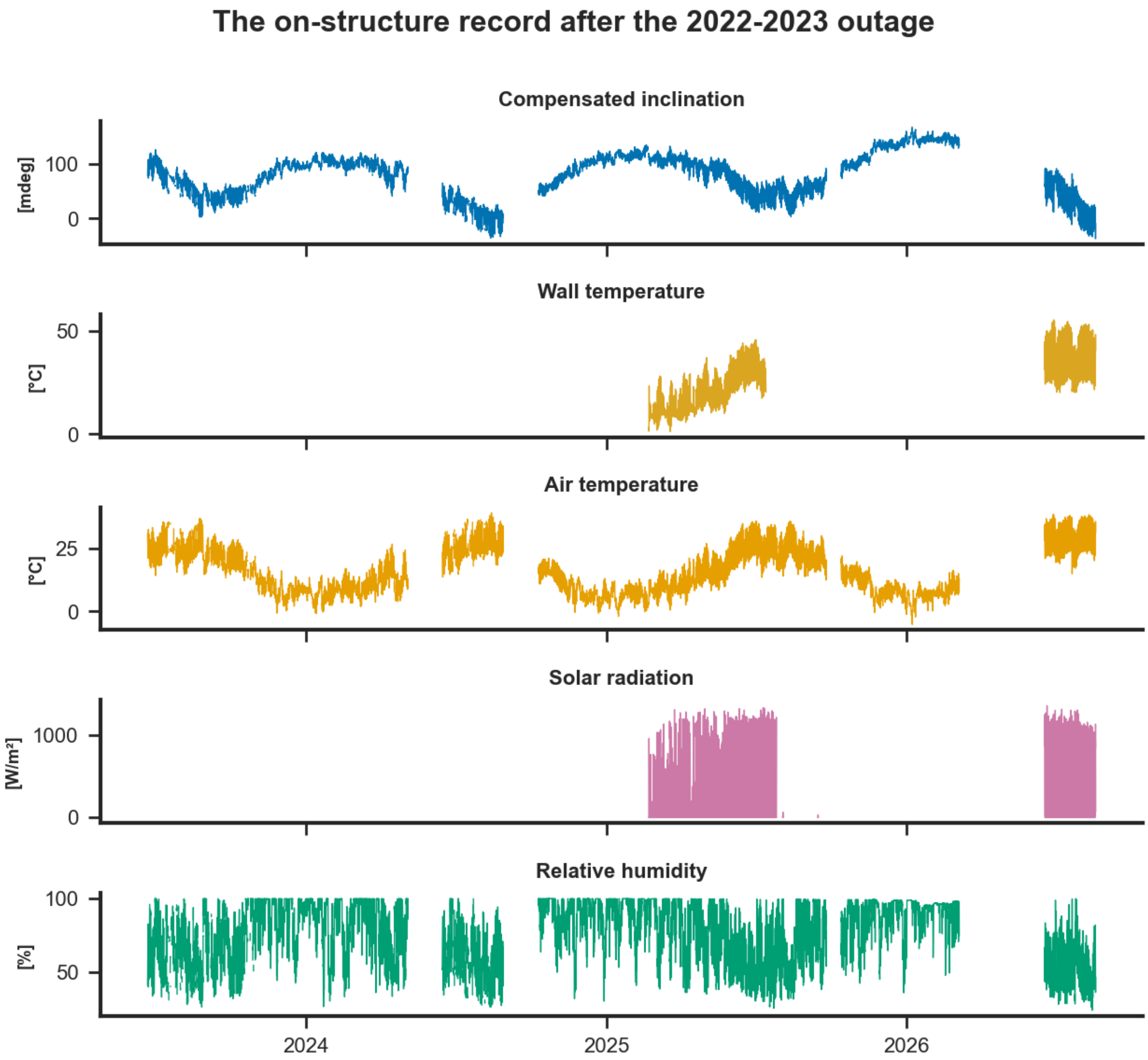

Figure 1: Every variable the instrument package on the wall records, over the window this study works on, on one shared clock. Nothing is interpolated and nothing is dropped before drawing: a gap in the record is a gap in the figure. Inclination, air temperature and relative humidity run from the resumption of recording in June 2023; wall temperature and solar radiation begin only with the package installed on 21 February 2025, and are themselves interrupted after it. The package also records its own supply voltage, which is instrument housekeeping rather than a measurement of the wall or of its environment, and is not drawn here.